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Situated visualization towards manufacturing maintenance training: Scoping review, design and user study☆

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ABSTRACT

Immersive technologies offer advantages for the visualization of and interaction with complex setups within manufacturing maintenance processes. The present work catalogs different applications of AR/VR in manufacturing maintenance practices as an extended version of a workshop paper presented at the International Workshop on eXtended Reality for Industrial and Occupational Supports (XRIOS). Through a scoping review in three computing and engineering digital libraries, we outline the key attributes of immersive solutions (N_p = 115) for industrial maintenance, categorizing functional prototypes with ten parameters related to interaction, visualization, and research methods. Moreover, we conducted a workshop with three manufacturing experts discussing the future of maintenance interfaces. By bringing forth their recommendations and insights, we targeted a key training challenge in maintenance. We designed and implemented a situated visualization prototype for a safety-critical procedure with real-time, in-depth, spatially relevant instructions. We compared the effects of 2D labels and 3D ghosts in a VR-simulated AR environment. In a preliminary between-subjects evaluation study (N_u = 24), we measured usability, workload, simulator sickness, completion time, and delayed recall. Although we did not find statistically significant differences between conditions, 3D ghosts showed slightly lower perceived workload and discomfort levels, along with shorter completion times. On the other hand, 2D labels produced higher usability. Overall, we contribute by a mapping out the state of the art and discovering knowledge gaps within immersive maintenance, presenting a design and preliminary user study that adheres to our recommendations.

1. Introduction

Industrial maintenance often anticipates machine failures through an evaluation of a machine's current condition and historical data to optimize the timing of maintenance activities [1]. Predictive maintenance, along with other similar methods, are sophisticated maintenance approaches used in many engineering fields, playing a crucial role in ensuring structural safety, and decreasing maintenance cost and time [2]. Currently, industrial maintenance is often comprised of two key aspects: the 'prediction' aspect focuses on forecasting machine failure and evaluating its remaining useful life [3,4], while the 'maintenance' aspect includes the decision-making strategies [5] and the actual maintenance and repair efforts carried out by operators and technicians.

Augmented and Virtual Reality (AR/VR) technologies have been widely used in many engineering fields, due to their significant advantages in visualization [6–8]. A recent review indicated that AR/VR technologies have been one of the four most employed maintenance solutions in industrial settings since 2019 [9], along with predictive maintenance, Artificial Intelligence (AI) methods, and digital twinning. These implementations aim to create a virtual replica of the physical assets by integrating the prognostic results and maintenance strategies into visualization models and algorithmic data processing [10–13]. As a consequence, industrial maintenance operations can be significantly

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streamlined and enhanced [7]. This improvement is evident from three perspectives: remote training, on-site operations, and cross-reality collaboration.

For remote training, VR-based training modules can provide maintenance staff with realistic simulations of equipment and scenarios, improving their skills and readiness for actual maintenance activities [14,15]. During on-site operations, one clear advantage for AR solutions is that maintainers are able to see machinery data that is spatially situated where it is most relevant on the factory floor, with the implementation of situated visualization (SV) [16,17]. Finally, combining these technologies can facilitate remote assistance by enabling experts (immersed in a VR digital twin) to guide on-site technicians (using a situated AR visualization) through complex maintenance procedures under cross-reality collaboration [18,19]. In return, this can reduce the need for physical presence and travel costs.

The use of AR/VR applications in maintenance and manufacturing has been extensively studied and implemented across various industrial fields. There have been substantial reviews published on these broader topics [6,7,20]. Under a narrower context, the present work intends to offer an updated overview of the intersection of AR/VR and state-of-the-art industrial maintenance research, focusing our efforts on the solutions presented in the last 6 years and filling the gap in time left since these works were published [6,7].

This work extends a previous workshop paper [21] presented at IEEE VR 2025, as part of the XRIOS workshop.[1] We contribute to the previous scoping review with broader search terms and adding a new target journal in the scope, as detailed in Section 2. Additionally, we introduce a novel immersive training design (Section 6) and a preliminary user study (Section 7), putting into practice the previously reported expert recommendations. This study is conducted under an industrial–academic partnership between KTH (Royal Institute Of Technology), Digital Futures, and AstraZeneca.[2]

By surveying and cataloging (Section 2) papers that demonstrate or analyze implemented systems and prototypes within this specific community, we aim to offer a categorization of its state of the art (Section 3), exposing research gaps (Section 4), prior to iterating with our own stakeholders (Section 5), designing (Section 6), and testing (Section 7) our own maintenance training solution. We later present our results (Section 7.4) and discuss our insights into their meaning (Section 8), prior to concluding our work (Section 9).

2. Review methodology

Based on the PRISMA [23] systematic review method, two authors (EJP and RG) conducted an exploratory search of the last six years (2020–2025), complementing the gap left since the publication of similar reviews [6,7], which explored the use of AR/VR applications in maintenance and manufacturing. Under a narrower context, our work offers an updated overview of the intersection of immersive technologies and industrial maintenance research.

The following search term was used to retrieve the initial records: *"predictive maintenance" OR "productive maintenance" OR "preventive maintenance"* AND (*"augmented reality"* OR *"virtual reality"* OR "immersive" OR "mixed reality" OR "extended reality").

Given the interest of our research in implemented systems and prototypes, as opposed to purely theoretical or business-focused works, the focus of our search was on engineering and computing venues. Thus, we constrained the search to the IEEE-Xplore,[3] ACM digital library,[4] and ScienceDirect[5] search engines. Particularly under ScienceDirect,

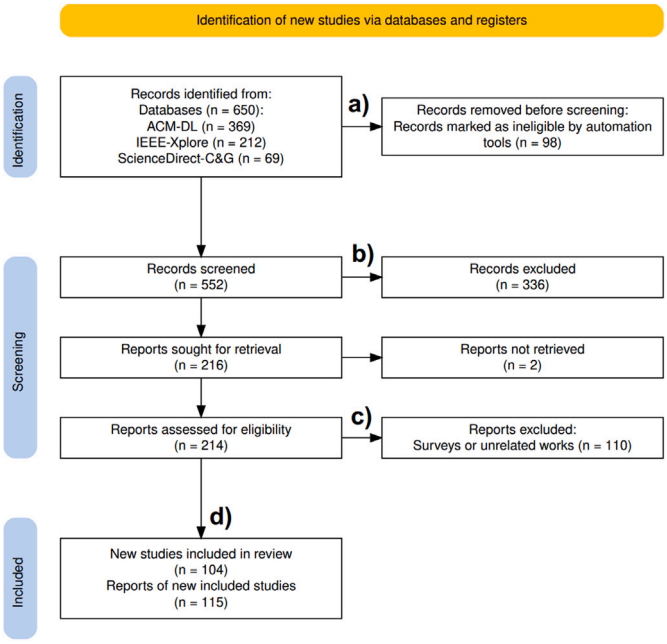


Fig. 1. PRISMA 2020-compliant flow diagram describing the search methods [22]. Inclusion and exclusion criteria: **(a)** 2020–2025 year range; **(b)** non peer-reviewed manuscripts; **(c)** surveys, literature reviews, and research papers unrelated to AR/VR technologies and industrial maintenance; **(d)** additional similar works included through snowballing. More details about each phase can be found in Section 2.

the Computers & Graphics journal[6] was exclusively targeted during our survey, due to its suitability to our research theme (immersive visualization and interaction) and given the fact that the ScienceDirect digital library encompasses research fields outside of our scope (computing and engineering).

In order to ascertain the relevancy of our results, a search entry was excluded in case it met any of the exclusion criteria described next and shown in Fig. 1: **(a)** published outside the 2020–2025 year range — removed during the identification phase; **(b)** non peer-reviewed manuscripts such as conference editorials or books — excluded in the abstract screening phase; **(c)** surveys, literature reviews, and any research papers unrelated to AR/VR technologies and industrial maintenance — all excluded in the full-text screening phase. Finally, **(d)** additional similar works were included in the review through snowballing the list of references and citations of each included article.

3. Survey results

By following the aforementioned research methods, we gathered a sample of N_p=115 highly relevant entries, including 87 conference papers and 28 journal articles.

Country. The set of countries in which the affiliations of the multiple authors of each paper are located. In summary, these can be grouped into Europe — 63 papers (46%); East Asia — 25 (18%); North America — 15 (11%); South East Asia — 9 (7%); South Asia — 7 (5%); Oceania — 5 (4%); and South America, West Asia, and North, West and Southern Africa, all with 4 papers each (3%).

Year. The distribution per year in which the papers were published, which is available in Fig. 2. A notable increase trend can be noticed

[1] https://sites.google.com/view/xrios

[2] digitalfutures.kth.se/research/industrial-postdoc-projects/smart-smart-predictive-maintenance-for-the-pharmaceutical-industry/

[3] https://ieeexplore.ieee.org

[4] https://www.dl.acm.org

[5] https://www.sciencedirect.com/

[6] https://www.sciencedirect.com/journal/computers-and-graphics

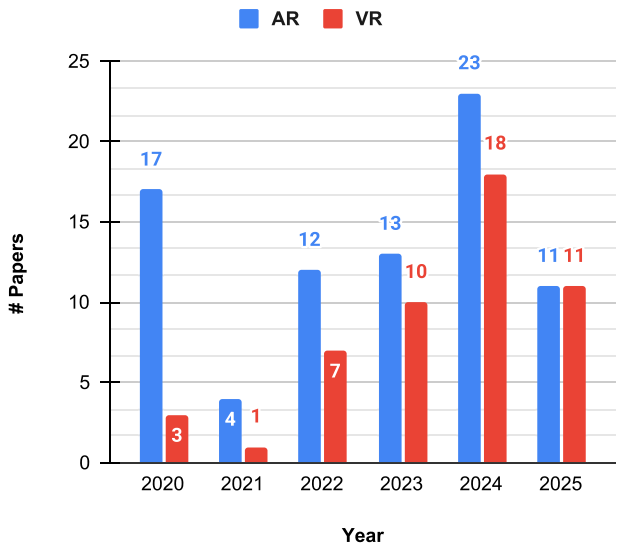


Fig. 2. Number of papers published from 2020 until October 2025. Notably, the sum of *AR* and *VR* columns does not represent the total amount of surveyed papers, as some works refer to both platforms.

Table 1
Examples of the industries referred to in the collected papers.

Industry	%	References
IT	14.7% (17)	[17,24–32] [33–38]
Transportation	14.7% (17)	[10,15,39–46] [47–53]
Construction and Urban Development	8.7% (10)	[16,54–62]
Energy	7.8% (9)	[63–71]
Additive and Subtractive Manufacturing	6.0% (7)	[72–78]
Aerospace and Defense	5.2% (6)	[79–84]
Mining	2.6% (3)	[11,19,85]
Logistics	2.6% (3)	[86–88]
Healthcare	2.6% (3)	[89–91]
Paper	1.7% (2)	[92,93]
Food	1.7% (2)	[94,95]
Retail and Services	1.7% (2)	[96,97]
Generic, unspecified, or multi-use	28.7% (33)	[98–106] [107–114] [115–122] [123–130]

along the years, except for the year 2021, which may arguably be explained by the COVID-19 pandemic and its effects on user studies [131]. Notably, the latest year available (2025) is still ongoing during the submission of this study, which explains its numbers being lower than the previous year.

Industry. The specific industries the papers intend to contribute to with their proposed prototypes, which is available in Table 1. The industries that stand out the most are Information Technology (IT), Transportation (including car, rail, and maritime), Energy, and Construction and Urban Development, likely due to their higher investments in high-tech solutions.

To provide a relevant map of the available literature on the current topic, we categorized the reports through deductive coding regarding the following ten attributes:

(1) Paradigm. The immersive interface paradigms that the papers address (AR, VR, or both), which is available in Fig. 2 and Table 2. Notably, the amount of AR papers has always been higher than VR, which is arguably understandable due to its *in situ* use.

(2) Prototype platform. Which type of device the prototypes are meant for — e.g. Head-Mounted Displays (HMD) or hand-held devices —, which is available in Table 2. The amount of prototypes using handheld platforms (e.g. smartphones and tablets — 16; 13.9%) is less than half of its HMD counterpart (e.g. AR and VR headsets — 50; 43.4%). This trend in hands-free interaction is expected due to the recent improvements in see-through and pass-through devices (e.g. 31; 26.9% used the HoloLens[7] series, and 18; 15.6% used the Meta Quest[8] series, respectively). In the cases where no prototype was implemented in the study, that paper was not categorized in the remaining attributes.

(3) Collaboration. Whether the prototypes are meant for a single user, or for multiple users in collaboration (single-user, multi-user). If multi-user, following Ruddle's [136] definition (symmetric: different users access the system from the same, or equivalent, platforms or paradigms; asymmetric: different users access the system from different, or not equivalent, platforms or paradigms), we also catalog the symmetry of the collaboration between users. These are available in Table 2. The majority of the prototypes only allowed a single user, with only 19 (16.5%) allowing multiple users to interact with one another via these different platforms. Only 8 studies (6.9%) implemented a symmetric collaboration scenario [54,98], with several of the asymmetric implementations (11; 9.5%) only allowing a remote person to call-in via voice or video [19,24,25,39,99], rather than providing a remote digital twin in VR. Given the benefits of cross-reality and remote collaboration in industrial practices [18], this indicates a highly valuable gap left unexplored.

(4) Liveness of the scenario:. Whether the prototypes are meant to be used in live operation scenarios (allowing for real-time data) or for training/education purposes (data is not necessarily live), which is available in Table 2. A considerable amount of the prototypes focused exclusively on live operation scenarios (40; 34.7%), with only a few exclusively pursuing training or education purposes (21; 18.2%), while the remainder encompassed both scenarios (21; 18.2%).

(5) Access location. Whether the prototypes are meant to be operated exclusively *in situ* (local only), or from anywhere (remote), which is available in Table 2. The majority of the prototypes exclusively aimed for the systems to run locally at their facilities (44; 38.2%), as opposed to also including remote access (37; 32.1%). This is expected since workers usually need to perform physical maintenance in the machines. Notably, remote access does not imply the absence of local access, so this can be seen as an extra feature.

(6) Multimodal feedback. Besides visual, we explored which modalities of feedback the prototypes are able to output to the users (e.g. audio or haptics), which is available in Table 2. Notably, 22 of the prototypes surveyed (19.1%) offered some sort of audio output integrated into their immersive experience, usually as part of their video content. Meanwhile, 5 papers (4.3%) addressed the use of haptics, with 1 paper (0.8%) surprisingly making use of olfactory feedback [105]. This low adoption of non-visual feedback suggests most of these systems have very low accessibility to visually impaired users [137,138].

[7] https://www.microsoft.com/en-us/hololens/
[8] https://www.meta.com/se/en/quest/

Table 2
Categorization of the papers analyzed in this scoping review.

Category	Value	%	References
(1) Paradigm	Augmented Reality	53.9% (62)	[15,17,19,24,25,27,39–41,54,55,57,72,79,80,85,93,96,99,101,106] [29,32,33,38,44,47,61,64,66,86–88,92,107,109–112,117,118] [34,37,52,53,70,76–78,91,102,103,105,108,123–127,129] [132,133]
	Virtual Reality	30.4% (35)	[11,30,45,50,51,58,63,65,74,75,81,82,94,98,104,114,116,128,130] [31,35,36,46,48,49,68,69,83,84,89,90,97,119–121]
	Both	13.9% (16)	[16,26,28,40,42,56,59,60,62,67,71,73,95,100,113,115]
(2) Platform	Head-Mounted Device	43.4% (50)	[11,16,17,24,25,39–41,54,57,58,63,64,85,99–101,107,110,111] [29,30,34,44,51–53,65,77,78,89,91,97,105,118,125–129] [31,35,36,62,67,69,90,95,120,122]
	Handheld Device	13.9% (16)	[15,19,37,38,61,66,72,76,86,92,93,106,108,109,133,134]
(3) Collaboration	Asymmetric	9.5% (11)	[19,24,25,30,37,39,53,69,78,99,135]
	Symmetric	6.9% (8)	[34,54,62,76,91,98,98,128]
(4) Liveness of the Scenario	Live Operation	34.7% (41)	[10,11,16,17,19,29,30,34,53,57–59,63,66,67,72,73,78,92,93] [31,36,44,45,75,97,104–109,111,113–116,119,126,127]
	Both	18.2% (21)	[24,25,39–41,52,54,61,65,69,76,85,86,91,99–101,110,117,118] [134]
	Training	18.2% (21)	[15,35,37,38,51,62,64,77,89,90,94,95,98,98,120,122,125,129,132,133] [128]
(5) Access Location	Local Only	38.2% (44)	[11,16,17,19,24,25,39–41,54,85,86,92,93,99–101,106,107,109] [35,36,38,44,52,61,65,66,77,105,108,110,117,119,122,125,126,129,132,133] [62,89,120,128]
	Remote	32.1% (37)	[10,15,16,29,30,34,57–59,63,64,72,73,98,100,111,113–115,118] [31,37,45,51,53,67,69,76,78,90,91,95,97,104,114,116,127]
(6) Multimodal Feedback	Audio	19.1% (22)	[19,28,35,37–39,53,61,76–78,85,89,90,95,98,101,108,110,117] [122,128]
	Haptics	4.3% (5)	[16,89,91,107,128]
	Olfactory	0.8% (1)	[105]
(7) Input Modes Available	Gestures	21.7% (25)	[16,17,29,30,36,40,41,54,57,66,69,77,99,100,107,111,120,122,126,129] [52,67,89,90,95]
	Touchscreen	14.7% (17)	[15,19,37,38,53,64,72,76,85,86,92,93,95,106,109,129,133]
	VR controllers	11.3% (13)	[16,29–31,51,58,62,63,65,69,97,120,128]
	Gaze	7.8% (9)	[17,40,41,52,95,99,107,111,125]
	Voice	7.8% (9)	[39–41,53,54,61,67,78,99]
(8) Situated Content	Animation	24.3% (28)	[10,11,16,19,40,41,52,59,78,85,86,92,97,100,106,109,110,115,120,132] [53,89–91,104,116,122,133]
	Data Visualization	20.0% (23)	[11,16,17,19,24,38,39,62,63,66,85,86,92,93,97,100,106,110,116,126] [65,69,122]
(9) Use of AI Methods		30.4% (35)	[11,19,25,34,40,41,57,63,73,92,99,100,104,107,109,110,113–115,118] [44,45,61,65,67,69,75,89,97,116,117,119,125,129,132]
(10) User Study		29.5% (34)	[15–17,19,30,37,38,40,41,52,54,64,72,78,85,91,98,117,125,132] [35,36,51,62,67,76,77,89,97,108,120,126,128,129]

(7) Input modes available. Which modalities of interaction the prototypes are implemented to accept as user input (e.g. gaze, gestures, voice, VR controllers, and/or touch), which is available in Table 2. Out of the papers that specified the input modes available, 25 included gestures or hand tracking (21.7%), 17 used regular touchscreen interaction (14.7%), 13 made use of VR controllers (11.3%), 9 made use of gaze interaction or eye tracking (7.8%), and 9 included some sort of speech interaction (7.8%).

(8) Situated content. Whether the augmented contents in the prototypes are presented in a situated form [139], and if so, which form of content (e.g. animation and/or data), which is available in Table 2. Out of the prototypes surveyed, 28 made use of situated animations (annotations,

3D models or videos; 24.3%), and 23 included some sort of situated data visualization (20%), with 12 implementing both modes (10.4%). Given the benefits of SV in industrial practices [16], these low numbers suggest another valuable gap to be further explored.

(9) Use of AI methods. Whether the prototypes made use of any AI methods, which is available in Table 2. In total, 35 papers (30.4%) disclosed making use of some AI model, although most of these did not specify what algorithms were used, other than reporting the use of "Machine Learning". This lack of clarity demonstrates a need for more reproducible studies to be conducted in this community.

(10) User study. Whether the papers included results from a user study, which is available in Table 2. Slightly over half of the total amount of papers surveyed (50.4%) did not present any results at all. Out of the 57 papers that reported any kind of result — either quantitative or qualitative —, 34 reported results from an original user experiment (29.5%). The remainder reported on system performance metrics (19; 16.5%), and pre-study survey results (3; 2.6%). Considering these papers are published in academic venues, this lack of experimental results demonstrates a concerning need for further experiments to be conducted within the AR/VR and industrial maintenance intersection.

4. Review discussion

Throughout the surveyed results, we notice a geographical distribution highlighting Europe as a leading contributor with almost half of the total papers, followed by East Asia and North America. The temporal analysis indicated a growing interest in the subject over time, with AR consistently being more focused in industrial maintenance immersive applications, though the balance could be shifting in favor of VR. Further insights revealed trends regarding platform usage, with HMDs having dominated over traditional handheld devices in the last 6 years, attributed to the recent advancements in AR/VR hardware. This finding goes against a recent survey by Fernández-Moyano et al. [140], although the authors acknowledge that this is to be expected in contexts where hands-free operation is essential, such as maintenance and assembly [140].

The industry focus shows a strong presence in sectors like construction, transportation, and IT, possibly due to high-tech investments in these fields. The lack of previous efforts within the pharmaceutical industry is rather surprising. This is either representative of a major research gap or confidentiality protocols within these companies. Regardless of the reason, we believe our project's focus on this type of research is rather pertinent and timely.

The collaborative setups explored suggest a gap in the exploration of multi-user environments, especially in symmetric scenarios, which are sparsely implemented despite the potential benefits for remote and cross-reality collaboration in industrial contexts. The current low adoption rates of single-user environments in the industry are likely the cause of this. It is possible that this might change in the near future, due to the evident ratio of handheld to head-mounted device prototypes shown in Table 2, indicating an expected trend in these industrial applications for the platform focus to shift from current smartphones to immersive and wearable interfaces, especially in maintenance and assembly tasks, where hands-free operation is essential [140]. This underexplored area presents an opportunity for future research.

Most of the prototypes were designed for live operational scenarios, with a third of those aimed at supporting educational contexts, reflecting a significant interest in real-time applications. Additionally, a majority of the prototypes were designed for local, on-site use, emphasizing the practical need for workers to interact with physical machines, though close to half incorporated remote access as an extra feature. Regarding multimodal feedback, the lack of non-visual outputs like audio or haptic feedback indicates limited accessibility for users with visual impairments, a gap that future research should address.

In terms of user interaction, touchscreens were the most commonly used input mode, with novel alternatives, considered more natural — e.g. gestures and gaze — being underrepresented, highlighting a minimal diversity in interface design. Situated virtual content appears frequently, though there is room for more integration of both data and animation to enhance industrial applications. AI was incorporated into some of the prototypes, though details on the algorithms used were often vague, indicating a need for more transparent and reproducible research. Notably, user studies were conducted in some of the papers, but the majority lacked any type of result analyses, pointing to a broader need for more experimental validation in this field.

5. Formative study

Subsequently, we conducted a workshop by presenting the summary of an earlier version (see Guarese et al. [21] for more details) of this scoping review to three experts in manufacturing. Each person performed complementary roles in the field of manufacturing, ranging from technical maintenance (E1, E3), hands-on machine operation (E2, E3), business innovation and IT (E1), and trainee instruction (E2, E3). As the review did not show enough evidence that the surveyed prototypes successfully scale into real-world manufacturing scenarios, we believe that more focus needs to be put into translating the results into applied solutions.

5.1. Participants

Within the context of the manufacturing industry, the invited manufacturing experts held the following roles:

E1: Senior Business Developer in Technical Development and Innovation (1 year). Formerly an IT Project Manager/Business Analyst (2 years), a System Maintenance Manager (3 years), a Service Engineer (3 years), and a Line Technician (5 years).

E2: Machine Operator (7 years) and Instructor for the Good Manufacturing Practice (GMP) [141] program (2 years). Formerly a first-line manager (1 year).

E3: Trainer in the Total Productive Maintenance (TPM) [9] academy (1 year). Formerly a Production Support Manager (2 years), a first-line manager (5 years), Business Developer and Production Technician (2 years), and a Machine Operator (7 years).

5.2. Qualitative feedback

During the aforementioned workshop, we introduced the experts to the goals of our predictive maintenance project, and presented them with a summary of the findings of this scoping review. Finally, we conducted a semi-structured interview with each one, asking for their feedback on which scenarios of their current work could make better use of these technologies. While the discussion prompts were deductively framed around the findings from the scoping review, the themes that emerged from the experts' responses were analyzed inductively. From this process, we identified four recurring opportunity areas, namely Operator Assistant, Remote Training, Multimodal Interface, and Situated Interface, which reflected shared needs and ideas across participants. These themes later informed the design of our prototypes, serving as foundations for mapping the expert feedback into applied design directions.

Operator assistant. Expert E1 suggested the training of new operators can be highly accelerated by using AI-powered Conversational Agents. They motivated this by describing the large number of documents describing maintenance procedures they need to read through and memorize. One example they provided was a changeover (the process of converting a line or machine from running one product to a new one), which had over 600 steps. They suggested that a context-aware or conversational assistant could parse the procedure information and describe it to the worker when it is relevant.

Remote training. Experts E1 and E3 valued the aspect of game-like environments for the training of maintenance procedures, such as troubleshooting. In their opinion, machine breakdowns are often caused by humans due to inexperience. By using VR, E1 said users should experience dismantling a machine into its small unique pieces, which would be unrealistic with the physical machines, due to time and cost. E3 believed this sort of remote gamified experience to be the ideal setting to explore within the TPM academy [9], which serves to train the staff to perform operator and specialist maintenance activities collectively. According to E3, TPM focuses on building a collaborative work mindset between different types of manufacturing professionals. They believed using this between classes could highly shorten the time it takes to qualify these professionals, especially by allowing remote and local people to collaborate synchronously, under cross-reality collaboration [18].

Multimodal interface:. Expert E1 suggested the need to focus on multimodal interfaces, especially audio and vibrotactile feedback [101], to represent what you hear and feel coming from the physical machines. When working as a line technician, they heavily relied on these senses while troubleshooting, since mechanical problems are often caught early with unexpected noise and vibration, under predictive maintenance. By replicating these problems during training, under multimodal feedback, E1 believes workers are likely to develop these sensing skills and start creating production value much earlier in their training.

Situated interface. E1 and E2 had previously explored the integration of video instructions in a Heads-Up Display RealWear[9] headset, as a substitute for the current paper-based documentation on maintenance procedures. At the time, videos were chosen as they did not have access to any spatially aware hardware. By focusing on the list of competencies a worker needs to have in order to be fully certified to handle a machine, they found it much more valuable to train new workers using videos due to their visual and motion description, and capacity to control their own learning. Especially since their synchronous time with instructors is very limited, and their access to certain restricted areas is not possible prior to acquiring the necessary certifications, due to safety, health, and environmental awareness reasons. E2 clarified that the older hardware they used did not offer the immersion and fidelity levels they were looking for, so the efforts for live AR instructions were discontinued then. Given our presentation, however, all three experts reported foreseeing much value in SV through AR, both for training, such as during the GMP program [141], and in live operation scenarios as well, such as a changeover, due to its spatial awareness, but especially since the user is free to use both hands at all times. E1 pointed out that their major concern was with how cumbersome current AR devices are, which could affect the willingness of people to wear them during their tasks. They added that this would be highly alleviated once slim and lighter devices such as the Meta Orion glasses[10] were commercially released.

5.3. Iterative process

The workshop surfaced several promising directions for integrated interface development, including conversational operator assistants (which is explored in a parallel study [117]), remote gamified training, and SV instructions. From these, accelerating operator onboarding emerged as a particularly valuable initial focus area, aligning with the company's goal of reducing the time required for workers to achieve independence while minimizing reliance on limited instructor resources. During follow-up discussions, expert E2 emphasized that many safety-critical procedures are currently being passed informally between experienced staff, creating inconsistencies in training. Much of this tacit knowledge [142] is undocumented as, in most cases, it is considered "common sense" by senior employees, according to E3; however, they agree it is critical for novices to absorb this knowledge, and there are currently no formalized protocols for it. This highlighted the need for tools that could externalize such practices into *standardized*, *step-by-step instructions*, an effort led in the past by E2, but discontinued due to lack of resources.

As discussed in the interviews, situated interfaces, implemented through immersive technologies, offered a natural solution for this [139]. Augmented and Virtual Reality (AR/VR) enable instructions to spatially and timely appear directly *where* and *when* actions must be performed — either on real equipment (AR) or in realistic digital replicas of it (VR). Such approaches have been shown to reduce cognitive load and errors while enhancing training outcomes [16,17].

Operating in a highly regulated environment introduced several practical constraints that shaped the choice of medium and implementation, making constant stakeholder feedback essential. Access to live production machinery was restricted, and according to E1 and E2, the available documentation for task guidance was largely text-based, with minimal audiovisual materials. Building a proof of concept on active machinery would have also required extensive operator supervision, which had been infeasible as stated by E2. To overcome this, E3 provided an out-of-service machine that could be safely examined and digitally replicated, conducting a TPM training session with three of the authors about its functioning and maintenance procedures. During this session, E3 suggested a critical maintenance safety procedure could be useful as a test bed for the initial evaluation: Lock Out, Tag Out, Try Out (LoToTo), which is further detailed in Section 6.2.

While AR was initially considered, its implementation in industrial environments is often difficult due to safety concerns, and workplace limitations [143]. In our case, no formal documentation was available to describe the labeling machine's operation (see Section 6.5). Additionally, access to the in-line system was restricted due to safety and malfunction risks.

VR was therefore adopted as a practical medium for building and testing. Simulating training in VR is particularly suitable for safety-critical contexts such as industrial manufacturing [144], allowing safe, and repeatable sessions [143]. E3 also reported that VR is already being used by the company to deliver manual training at specific facilities, where trainees are able to safely repeat sessions at their own pace without requiring live equipment or instructor time. These existing VR training materials, however, had never been tested for their effectiveness in any metric at the company level, according to E3, which urged a pilot test to be implemented. This choice was also supported by a prior study showing that VR-based AR simulations effectively replicate real-world AR usability and task performance, making VR a suitable environment for evaluating AR-based SV [145].

Thus, we implemented our training prototype in a VR-simulated AR environment, which supported iterative testing of SV strategies without disrupting factory operations. At each step of implementation, stakeholder feedback was a primary driver of development. More about our process can be found in Vasiliu and BagheriFard [146]. This iterative refinement clarified both opportunities and constraints, informing our choice of target procedure and design approach.

6. Training prototype design

Based on our scoping review, research gap, and expert workshop, we designed a prototype for manufacturing training that provides spatially relevant, in-situ instructions for guiding users through safety-critical maintenance procedures. Focusing on training offered additional advantages: it created a non-critical, risk-free environment and allowed testing even with external participants — particularly valuable given the limited availability of experts. This section presents the training prototype in detail, including the design rationale, task scenario, visualization strategies, scene structure, and implementation setup that shaped the training prototype.

[9] realwear.com/

[10] about.meta.com/realitylabs/orion

6.1. Design goals and motivation

The training prototype was developed to examine how SV strategies can support users in safety-critical procedures. Our aim was to provide realistic, spatially grounded guidance, delivered through standardized, step-by-step instructions. To achieve this, we implemented two visualization conditions — 2D labels and 3D ghosts — adapted from Lee et al. [139].

We chose to implement the prototype in a VR-simulated AR environment to enable controlled, risk-free evaluation without relying on access to live machinery or instructor supervision, a common approach for user testing AR applications in recent years [143]. This approach not only ensured ecological validity for future AR deployment, but also allowed systematic testing of design alternatives under safe and repeatable conditions. The system was developed in Unity 3D and deployed on Meta Quest Pro head-mounted displays (HMDs), balancing immersion with technical feasibility and practical deployability in industrial training contexts (see more details about the implementation in [147]).

6.2. Task scenario

For testing the prototype, we collaborated with company experts to select the LoToTo safety procedure. LoToTo is a standardized protocol for safely isolating machines from energy sources and ensuring they are not reactivated before maintenance or repair work is complete. This procedure is critical for safeguarding employees from the unexpected release of hazardous energy during maintenance activities [148]. The task was selected because it is a safety-critical industrial procedure that requires clear instructional guidance to ensure compliance and understanding.

6.3. Visualization proposal

Since LoToTo is a safety procedure, it was important that trainees fully understood each step and performed it correctly. At the time of our study, LoToTo training was typically delivered in person and relied on experienced instructors. This approach might result in inconsistencies in how new employees performed the tasks and could make it difficult to standardize evaluations or ensure proper skill acquisition. Therefore, the visualization we implemented needed to be accurate and realistic enough to be easily translated to real-world practice, while also providing consistent guidance that could be applied reliably across trainees.

We chose to reproduce the LoToTo procedure using motion-captured data from an instructor's hand movements to demonstrate each step effectively. To ensure that the participants followed the essential safety steps — locking out machinery, tagging their names on the equipment, and verifying (trying out) the isolation from any energy sources — the study simulated a virtual operator performing the LoToTo process in VR under two visualization conditions.

At the beginning of the training, users were shown a simple welcoming panel; after clicking *Start*, they were guided through the three main LoToTo steps in sequence, using the labeling machine (Section 6.5) and the interactive VR objects (Section 6.4.2).

Step 1: LockOut — Participants need to identify relevant energy sources and lock them out securely. This includes opening the fuse box door, switching off the electricity, closing the door, locking it with the key, and securing the key.

Step 2: TagOut — Participants need to attach an identification tag to the locked equipment to inform others and indicate the responsible person. Activities include signing the tag and attaching it to the locked fuse box.

Step 3: TryOut — Participants need to verify that the equipment is completely locked out. Activities include confirming that the labeling machine is powered off and cannot be reactivated.

Each step was textually described on a separate instructional panel. In the end, a final completion panel confirmed the successful completion of the procedure.

6.4. Procedure and scene design

The training environment was structured into two sequential scenes: a tutorial scene and a main task scene based on the LoToTo procedure.

6.4.1. Tutorial scene

All participants began by interacting with a tutorial scene (similar to Fig. 3) designed to introduce the virtual environment and interaction mechanics. In this phase, users explored the virtual space freely, learned how to interact with interface elements — such as pressing buttons and navigating around the labeling machine (described in detail in Section 6.5) — and were encouraged to ask questions if anything was unclear. This scene did not include any instructions related to the LoToTo procedure. Detailed descriptions of the tools and objects used in the LoToTo process are provided in Section 6.4.2. In addition to those objects, the tutorial scene also included a small table with a start button used to power on the labeling machine.

The tutorial scene was designed to achieve two objectives. First, it ensured that all participants became familiar with the environment and interactions, which was particularly beneficial for those without prior experience in VR. Second, it prevented the time spent on exploration or clarification from being counted towards task completion time in the main study, allowing for a more accurate performance measurement.

6.4.2. LoToTo training scene

After completing the tutorial, participants were guided into the main LoToTo training scene (Fig. 3). Each participant experienced only one of the two SV conditions (2D or 3D). The LoToTo procedure was visually demonstrated step by step according to the assigned condition. Participants were expected to observe and replicate the steps using natural gestures in both cases. To evaluate the clarity and sufficiency of the visual instructions, participants were asked not to pose questions during this phase.

The virtual workspace contained key interactive assets that replicated real-world objects used in the LoToTo procedure: a fuse box for controlling the energy source, a key for locking the fuse box door, a pen for signing the identification tag, and the tag itself, which warned others that maintenance was in progress and identified the responsible person. These objects were placed on a table to mimic a natural setting. A working labeling machine (described in detail in Section 6.5) was also included in the scene and was simulated to power down when the trainee switched off the electricity.

This structured scene design directly compared the two SV patterns while ensuring a consistent user experience across conditions.

6.5. Apparatus and implementation

The VR training environment was developed in Unity 3D, using the Meta XR Interaction SDK, and was deployed on Meta Quest Pro HMDs to simulate AR interactions in a VR context.

To reflect a realistic maintenance scenario, a 3D model of an industrial labeling machine (Neri Marchesini BL400[11]) used at AstraZeneca was integrated into the training system, as shown in Fig. 4.

The 3D models for the environment, fuse box, key, pen, and tag were sourced from the Unity Asset Store and adapted to fit the training scenario. Tables were tailored from the Meta XR Interaction SDK sample scenes.

[11] https://intimac.it/en/used-machines/labelling-machines/labelling-machines-neri-marchesini-bl400/

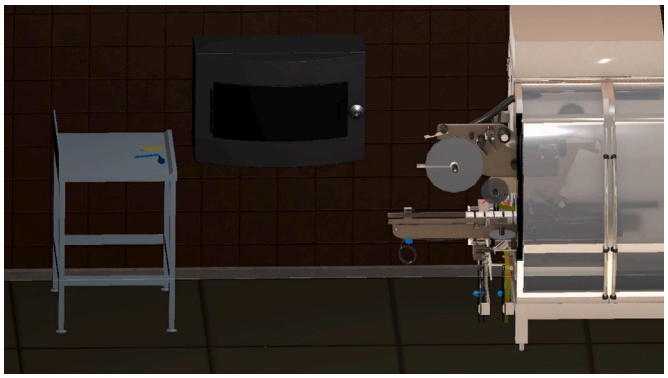


Fig. 3. LoToTo training scene, including a fuse box, key, pen, and tag placed on a table, and a working labeling machine.

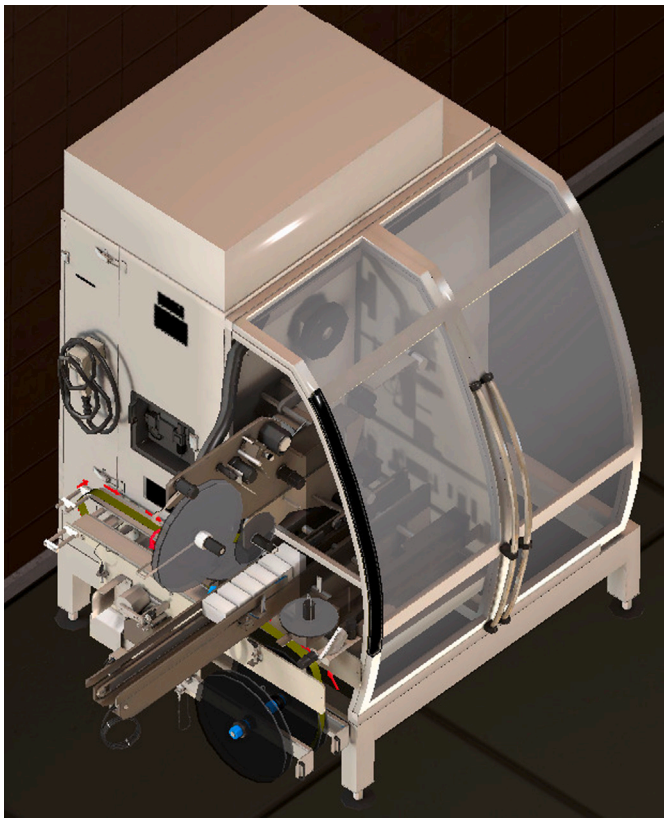


Fig. 4. 3D model of the labeling machine used in the training environment.

6.6. Apparatus and implementation

The scene, environment, and assets were prepared in Unity. We integrated the Meta XR Interaction SDK to enable object interaction, allowing trainees to open the fuse box door, switch off the power using the button, close and lock the door, and secure the key. They then signed the tag with a pen, attached it to the locked fuse box, and finally confirmed that the machine was fully shut down. These interactions used Grab/Rotate transformers, Poke interactors, Grabbable components, and custom scripts. Each step had an instructional UI panel introducing the required action. Scripts and shaders also rotated the machine's wheels and labels at the beginning of the simulation, imitating a running state that was then switched off via the button.

In both conditions, labels connected each task object to its corresponding UI text. Instructions were identical across conditions to provide step-by-step guidance and ensure a fair comparison, following best practices in controlled HCI studies [149].

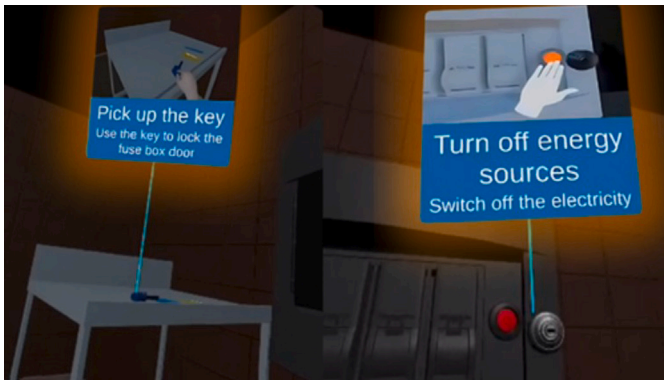


Fig. 5. 2D panel condition, showing UIs and video panels positioned in the task location, connecting with labels to the corresponding objects.

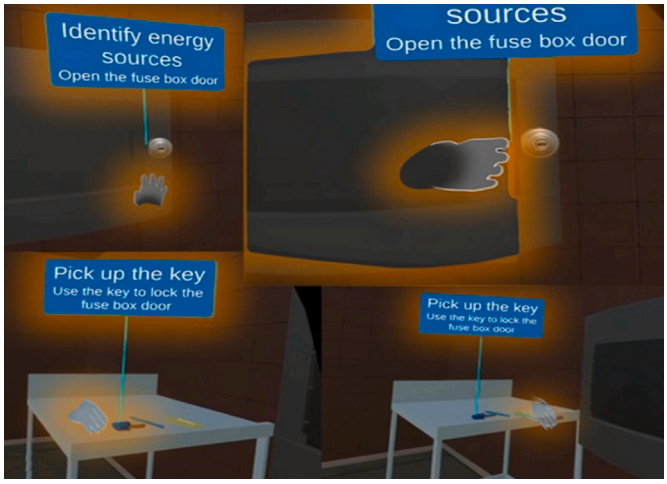


Fig. 6. 3D ghost condition showing text UIs and the ghost objects embedded in the task location.

6.6.1. 2D condition

In the 2D condition, spatially situated labels (as defined by Lee et al. [139]) displayed short video clips of each step of the LoToTo procedure, positioned precisely where the learner needed to execute them, as shown in Fig. 5.

To create these clips, the full LoToTo procedure was performed in VR using the Meta Quest Pro headset, and the headset's built-in recording feature was used to capture videos from first-person perspective. The recordings were then processed in the CapCut platform to crop and highlight the relevant actions. Final clips were integrated into the instructional panel using Unity's Video Player component, configured to render on dedicated textures within the scene.

6.6.2. 3D condition

The 3D condition placed a ghost (as defined by Lee et al. [139]) representation of the instructor's hands performing the same procedure in situ, along with ghosted versions of the objects they interacted with (all rendered translucent to avoid confusion), embedded in the scene [150], as shown in Fig. 6.

These animations were created using an animation controller, animation clips, and custom scripts to replicate the instructional sequence and synchronize it with object interactions in the virtual environment.

7. Preliminary user study

7.1. Study design

This section details the methodology used to evaluate the training prototype in a preliminary user study. We conducted a between-subjects experiment, using *visualization type* as the independent variable, under two levels: 2D labels vs. 3D ghosts. We measured its effects on dependent metrics such as task completion time, usability, workload, discomfort, and memory recall. Although exploratory, this protocol was designed to support reproducibility, enable clearer cause-and-effect interpretation between design patterns and outcomes, and reduce learning bias during analysis [151].

7.2. Participants

We recruited N_u=24 university students to simulate novice workers undergoing industrial safety training, ensuring that participants were unfamiliar with the LoToTo procedure to evaluate its learnability potential. Each participant completed the training scenario in only one SV condition, either 2D labels or 3D ghosts.

Convenience sampling was used due to time and accessibility constraints, which was appropriate for small-scale experimental research [149]. To mitigate potential bias, we recruited participants with different VR experiences and educational backgrounds. The participant allocation process was intentionally distributed to systematically balance the participants with and without prior VR experience and ensure fairness and comparability between the groups. This was achieved by selecting students from a technical program with VR experience and balancing them with participants recruited from other programs who had no prior VR experience. This resulted in each group having seven people with prior VR experience and five without. Collecting prior VR experience was essential because familiarity with immersive technology can influence outcomes. Less-experienced users often report lower usability and more nausea, whereas experienced users show the opposite [152,153].

7.3. Data collection

The following variables were collected to compare both conditions:

- **Task Completion Time:** A Unity script started timing when the first task UI was activated and stopped at the final UI, computing duration in seconds. Per-session logs were exported to CSV. We used these times to test whether the different visualizations affected procedural speed.
- **Usability (SUS):** Perceived usability was measured post-task using the System Usability Scale (SUS) [154]. Scores range from 0–100, with higher scores indicating better usability.
- **Workload (NASA-TLX):** Perceived workload was measured post-task using the NASA Task Load Index (NASA-TLX) [155]. Participants rated six dimensions (Mental, Physical, Temporal Demand, Performance, Effort, Frustration) on 10-point scales; we used the unweighted TLX for simplicity and reliability [156].
- **Discomfort (SSQ):** Discomfort was measured post-task using the Simulator Sickness Questionnaire (SSQ) [157]. Participants rated 16 symptoms; standard scoring yielded Nausea, Oculomotor, and Disorientation subscales.
- **Memory Recall:** To measure memory recall, participants were emailed a questionnaire two weeks after the VR session in which they were asked to recall and explain the three critical steps of the LoToTo procedure, as well as the rationale behind each step. To reduce pressure, participants were reminded that the goal was to evaluate the system, not their individual performance.

7.4. Results

All quantitative analyses and visualizations were conducted in Python. Following good practices on quantitative user research [158], we analyzed data from N_u=24 participants comparing 2D labels and 3D ghosts. Assumptions were checked with Shapiro–Wilk tests for normality and Levene's tests for homogeneity of variance. Parametric tests were used to verify whether paired comparisons were statistically significantly different ($p < .05$) when assumptions were met; otherwise, Wilcoxon rank-sum tests were applied.

The overall results did not present any statistically significant differences between the two conditions ($p \geq .05$). However, 3D ghosts showed slight advantages in perceived workload, physical comfort, and task completion times, whereas perceived usability was slightly higher in 2D labels. The visual summary of the results is in Fig. 7, and the detailed results are described below.

7.4.1. Task completion time

Median completion times are shown in Fig. 7, first row, first box plot; lower values indicate greater efficiency. The 3D condition ($Md = 121.72s$) was faster than 2D ($Md = 151.97s$), a 19.9% reduction.

Before running the tests, we checked assumptions; Shapiro–Wilk tests results did not show evidence of non-normality (2D: $W = 0.9144$, $p = 0.2428$; 3D: $W = 0.8948, p = 0.1359$), and Levene's test supported equal variances ($F = 0.0350, p = 0.8542$). An independent-samples t-test found no significant difference between conditions ($t = 0.6781, p = 0.5048$).

7.4.2. Usability

Median SUS scores are displayed in Fig. 7, first row, second box plot; higher values indicate better perceived usability. The 2D condition ($Md = 85.0$) showed slightly higher SUS scores than the 3D ($Md = 82.5$), a 2.9% difference. Both conditions were above the commonly accepted SUS benchmark of 68 [159].

Shapiro–Wilk test results indicated non-normality for the 3D group (2D: $W = 0.9129, p = 0.2327$; 3D: $W = 0.7838, p = 0.0062$). Given this, we used the Wilcoxon rank–sum test (Mann–Whitney U), which indicated no statistically significant difference between groups ($U = 88.00, p = 0.3686$).

An extreme outlier was identified in the 3D condition (SUS = 2.5/100).

7.4.3. Workload

Median NASA-TLX scores are shown in Fig. 7, first row, third box plot; reflecting participants' perceived task load; lower scores indicate better outcomes. Workload in the 3D condition ($Md = 3.42$) was marginally lower than in the 2D ($Md = 3.50$), a 2.4% reduction.

Shapiro–Wilk test results did not show evidence of non-normality (2D: $W = 0.8813, p = 0.0912$; 3D: $W = 0.8915, p = 0.1230$), and Levene's test indicated equal variances ($F = 0.5315, p = 0.4736$). Given that both assumptions were met, an independent-samples t-test found no statistically significant difference between conditions ($t = 0.5292, p = 0.6020$).

7.4.4. Discomfort

Simulator discomfort was assessed with the SSQ, where lower values indicate less discomfort and better outcomes. General SSQ in the 3D condition ($Md = 15$) was lower than 2D ($Md = 27$), a 44.4% reduction. SSQ scores also report three subscales: Nausea, Oculomotor strain, and Disorientation [157]. The overall SSQ results, along with its subscales, can be seen in Fig. 7, second row.

Nausea. Median nausea scores reflect symptoms such as stomach awareness, increased salivation, and general queasiness. Nausea in the 3D condition ($Md = 0.00$) was lower than in the 2D condition ($Md = 9.54$), a 100% reduction in the median.

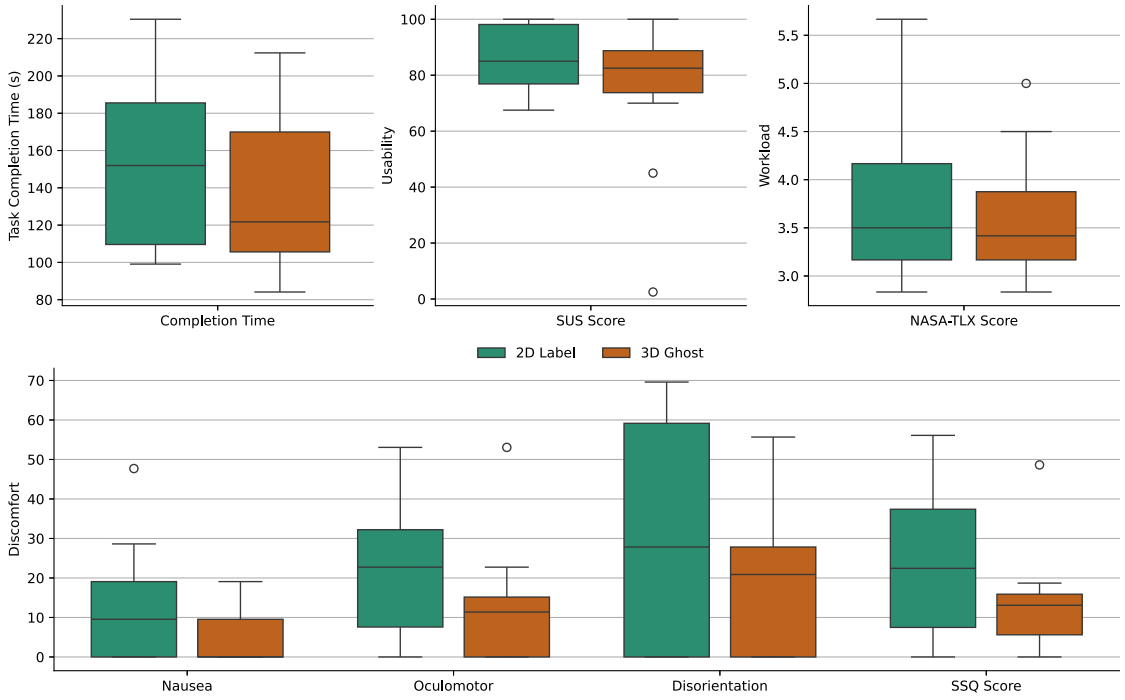


Fig. 7. Experimental results by condition.

Shapiro–Wilk test results indicated non-normality for the 3D group (2D: $W = 0.8770, p = 0.0803$; 3D: $W = 0.7321, p = 0.0017$). Accordingly, a Wilcoxon rank–sum test indicated no statistically significant difference between groups ($U = 96.5000, p = 0.1359$).

Oculomotor strain. Median oculomotor strain scores reflect symptoms such as eye strain, visual fatigue, and difficulty focusing. As shown, the 3D condition ($Md = 11.37$) was lower than the 2D condition ($Md = 22.74$), a 50% reduction.

Shapiro–Wilk test results indicated non-normality for the 3D group (2D: $W = 0.9345, p = 0.4297$; 3D: $W = 0.7720, p = 0.0046$). A Wilcoxon rank–sum test indicated no statistically significant difference ($U = 95.5000, p = 0.1580$).

Disorientation. Median disorientation scores reflect symptoms such as dizziness, balance disturbances, and spatial confusion. As shown, the 3D condition ($M = (Md = 20.88)$ was lower than the 2D condition (27.84), a 25% reduction.

Shapiro–Wilk test results indicated non-normality for both groups (2D: $W = 0.8328, p = 0.0227$; 3D: $W = 0.8493, p = 0.0360$). A Wilcoxon rank–sum test indicated no statistically significant difference ($U = 87.0000, p = 0.3830$).

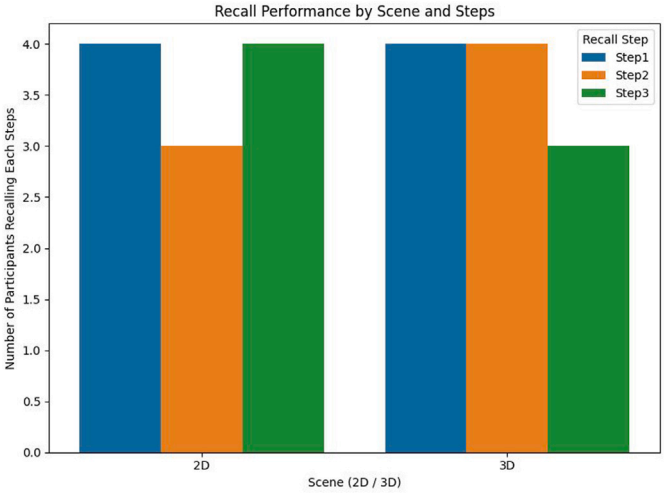


Fig. 8. Recall performance by scene and steps.

7.4.5. Memory recall

We sent follow-up emails with a brief recall task two weeks after the experiment. Only 4 of 12 participants per condition responded. Therefore, analyses used respondents only ($N = 4$ per condition), with non-responses excluded. The task comprised six open-ended questions — two per LoToTo step. Responses were scored 0–3 in total (1 point per correctly recalled step). As shown in Fig. 8, recall was similar across conditions overall. Step 1 was most frequently recalled. In the 2D condition, Step 2 had the lowest recall count, whereas in the 3D condition, Step 3 had the lowest.

8. Discussion

This section interprets the findings, evaluates the 2D labels and 3D ghosts, addresses the study's limitations, and proposes directions for future work.

This study explores how different SV techniques (2D videos vs. 3D ghosts) affect UX and task performance in industrial maintenance training. The primary finding is that both visualization techniques effectively guided users through the safety-critical LoToTo procedure by providing clear, spatially relevant guidance at the point of action.

8.1. Interpretation of results

High usability and manageable workload were reported across both conditions in our study. All participants completed the assigned task. Since both techniques were delivered in an immersive 3D environment, the 3D ghost guidance did not show a clear advantage over the 2D video overlays. A likely explanation is that instructions were anchored directly to the relevant objects, keeping them in view, which is a key principle of situated interfaces [160]. We did this by spatially anchoring guidance to the fuse box, key, and tag at the moment of need. This

suggests that linking guidance to the referent matters more than the dimensional format.

Accordingly, the results indicate no statistically significant differences between the two conditions across all outcome measures. However, for task performance, 3D guidance was faster in completion time than 2D. Memory recall appeared similar between conditions, but the limited number of follow-up responses prevents firm conclusions. In addition, for UX, both conditions achieved high usability scores; the 2D condition was slightly higher than the 3D. Perceived workload was low and nearly similar in both groups, with the 3D being lower than the 2D. Finally, the 3D condition consistently resulted in lower symptoms on all three SSQ subscales, suggesting that participants in the 3D condition experienced milder discomfort.

Our finding of no statistically significant differences between the 2D and 3D conditions for key metrics like task time and workload echoes recent findings in SV research. For instance, McGee et al. compared six different AR visualization techniques for hazard awareness and found no single option with a significant performance advantage, noting that user opinions on the best format varied widely [161]. Likewise, Aygün et al. reported no significant difference in perceived usability or workload between HMD and HHD interfaces for a collaborative assembly task, despite a clear preference for HMD due to its hands-free ergonomics [162]. Taken together, these studies and our results suggest that for SV guidance, clear, spatially relevant guidance matters more than the specific format (in our study, the dimensional format). As long as the visualization is placed in a meaningful location and is easy to understand, both 2D and 3D formats can be effective for task performance and UX.

This is especially informative relative to prior work on AR labels, which primarily addresses label placement, including the avoidance of clutter and leader-line crossings. For example, Zhou et al. propose a layout to help users find target labels faster in dense scenes [163]. Zhu-Tian et al. use deep reinforcement learning to optimize label placement for moving objects over time, balancing immediate readability with longer-term layout stability [164]. By deliberately setting aside the complex algorithmic challenges of label placement, our study focused on the dimensional format of the instructions. We employ a dynamic, task-dependent placement strategy where instructional labels are not fixed but appear sequentially, anchored to the specific object required for the current step. Once the user interacts with the object (e.g., grabs the tag), the label remains visible until that step is complete. This context-aware approach allows us to hold the placement strategy constant while varying the instructional format (2D vs. 3D). Our results suggest that when labels are presented in this clear, task-relevant manner, designers have considerable freedom to choose between 2D and 3D instructional formats without a major loss in performance or usability.

A gain in task speed and lower workload was expected with the 3D label, although the advantage we observed was non-significant. In this application, the 3D visualization provides a more direct, egocentric (first-person) representation through embedded visualization. By contrast, the 2D visualization is exocentric (third-person) and may require additional mental translation from the video feed to the actual environment, which could slow participants down [165]. It also aligns with ergonomic findings in related work. Aygün et al. reported a strong user preference for hands-free HMDs over HHDs in an assembly task because they were less physically demanding [162]. Likewise, Calepso et al. found that having to hold a tablet created usability issues during a physical task [160]. While our study used the same HMD hardware with hands-free interactions in both conditions, the 3D ghost visualization, which directly mirrors human action, may have created a more intuitive and less cognitively demanding mapping between the instruction and the physical task. This outcome also partly relates to findings from prior research [17], who reported that 3D visualizations can improve task speed but may also influence other aspects of performance.

Regarding usability, both conditions achieved high SUS scores, with mean values well above the benchmark of 68, indicating well-accepted designs [159]. This aligns with findings from [17], who reported that advanced visualization methods can achieve high usability ratings, suggesting that users adapt well to both forms of guidance in industrial and AR contexts. The slightly higher score in the 2D condition could be attributed to the familiarity of video interfaces. An extreme outlier was observed in the 3D condition (SUS = 2.5/100). The participant (P1) informally noted that the VR headset felt physically heavy and preferred to watch the tutorial on another platform. This indicates that hardware comfort can strongly affect usability ratings independent of the visualization itself. Independently, both conditions approach the threshold of 80, which is associated with excellent usability [159].

For workload, scores were low and nearly similar across both conditions. This indicates that the task requires only a manageable level of perceived workload, regardless of the visualization used. This aligns with observations by Guarese et al. [17], who noted that SV can enhance clarity and user understanding, potentially helping users perform tasks efficiently without excessive task load. Additionally, Zhou et al. showed that an optimized AR label layout significantly reduced workload compared with a conventional layout [163]. Overall, our results show that when guidance is clear and spatially relevant, the task can be completed with a manageable workload.

For discomfort, while earlier work by Guarese et al. did not report SSQ subscores specifically and reported only the overall score, they noted that well-designed SV techniques can reduce overall user discomfort [17]. Our findings align with this result. We suggest that the spatial clarity and intuitive guidance in 3D visualizations can contribute to lower physical discomfort symptoms. The egocentric nature of the 3D ghost may also help by creating a more direct and less cognitively demanding mapping between instruction and action than the exocentric 2D video.

Memory recall results appear similar in general, though the low number of follow-up responses makes it difficult to draw broader conclusions about how well participants retained the training content. Only a subset of participants (four per group) responded to the recall questions sent two weeks after the experiment. This can raise the risk of survivorship bias, which occurs when analyses focus only on cases that survived or succeeded, while overlooking those that did not, leading to an incomplete perspective [166]. It is possible that those who remembered the steps better were more likely to respond, while participants who could not recall the information may have chosen not to reply. As a result, the reported recall performance overestimates the true level of recall across all participants. In addition, using students without an industrial background likely reduced motivation to encode the safety procedure for long-term memory, unlike real-world trainees.

In conclusion, the study reveals that both visualization approaches are highly usable and effective for guiding users through a safety-critical maintenance procedure. While no statistically significant differences were found between the conditions, the results suggest that the primary design consideration is the clarity of the spatial link itself, rather than the specific dimensional format of the instruction.

The findings indicate that the small comfort and time advantages of the 3D condition may be particularly valuable in long-term or complex training scenarios where physical fatigue can become a significant issue, while the 2D condition received slightly higher usability scores. This supports the findings from McGee et al. which suggest that there may not be a single better visualization format for all users and recommend offering customization options, so that trainees can select the format they find most effective [161].

Therefore, when training is long or physically demanding, the small comfort and time advantages of 3D ghosts may be valuable; when usability, especially readability and a familiar UI, is the priority, 2D labels may be preferable. Overall, this indicates that designers have flexibility in choosing the dimensional format of instructional content based on context, task complexity, technical constraints, or user preferences. Our trends offer a clear guide: choose 3D ghosts when speed and comfort matter, especially in long or complex training; choose 2D video labels when familiarity and straightforward usability are the priority.

8.2. Limitations

The findings should be interpreted in light of several constraints. Firstly, the task used was somewhat simple, although of utmost importance to maintenance, and the relatively small sample size limited the statistical power of the results, reducing confidence in detecting differences between the two visualization conditions. This is particularly evident for memory recall, where the results were drawn from only a small subset of participants. This generates a risk of survivorship bias, hence, the primacy effect in recall should be regarded as anecdotal.

Moreover, although the study targeted novice operators, the reliance on student participants, rather than novice industrial trainees, may have constrained ecological validity and weakened the generalizability of the findings to real-world industrial settings. Unlike students, industrial trainees can differ in their prior exposure to machinery, safety protocols, and workplace practices — even at the novice level — which could affect not only task performance, but also how effectively procedures are recalled after training.

Furthermore, the study did not benchmark SV training against traditional instructional approaches (e.g., in-person or paper-based methods), which are still widely used in industry. Without such a benchmark, preferably as a longitudinal study, it remains unclear whether the observed benefits of SV approaches reflect genuine advantages over conventional practices or simply novelty effects.

The experiment was also conducted in a VR environment that simulated AR interactions with a virtual machine, rather than in a conventional AR setting on actual machinery, which may reduce external validity. These differences may have influenced the extent to which the findings transfer to operational environments. In addition, the study focused on a single maintenance task (LoToTo), with straightforward steps, therefore it is unclear how well the findings would generalize to other, more complex types of procedures.

Although the quantitative data provided detailed insights into task performance and SV effectiveness, the absence of qualitative data (e.g., post-task interviews) limited the understanding of participants' subjective experiences and interpretations [151].

8.3. Future work

Future research should address the current limitations and extend these findings in several directions. Increasing the sample size would improve statistical power and allow more reliable between-group comparisons. Recruiting novice industrial trainees in situ, rather than students, would further enhance ecological validity. Benchmarking SV training against common baselines (e.g., in-person or paper-based instruction) would help clarify the practical benefits of this technology, while evaluating a wider range of maintenance tasks would test generalizability across scenarios. Incorporating qualitative methods (e.g., interviews) could also provide deeper insights into user perceptions and design preferences [151].

Field evaluations with optical see-through AR on real machinery are warranted to ensure transfer to authentic industrial settings. In addition to completion time, future studies should assess performance metrics such as success rates, number of errors, and adherence to safety-critical steps [167]. Exploring SV patterns beyond labels and 3D ghosts — such as glyphs or contextual overlays [139] — may reveal combinations that better support efficiency, comprehension, and safety in industrial training.

Finally, the choice between 2D and 3D representations should be guided by contextual demands, including comfort, usability, task complexity, and session length. While 2D labels remain a practical option for simpler procedures and usability-focused designs, prioritizing 3D-based guidance may accelerate learning and reduce user discomfort in more complex scenarios. Continued refinement of these technologies has strong potential to enhance industrial training outcomes by improving both task performance and user experience.

9. Conclusion

This extended version of a previous XRIOS workshop paper [21] performed an expanded scoping review of immersive solutions within industrial maintenance tasks. We mapped and analyzed 115 papers, categorizing them in ten parameters related to their interaction, visualization, and research methods. This review informed the design, development, and investigation of how different SV designs — 2D labels versus 3D ghosts — affected UX and task performance in an AR-based training environment for industrial maintenance tasks. Labels and ghosts were chosen as the focus because they are a core pattern in SV, widely highlighted in recent research for their flexibility and potential to enhance AR guidance [139]. Through a preliminary controlled experiment with 24 participants, differences between conditions were not statistically significant across the measured outcomes. Both 2D and 3D showed high perceived usability and were generally well accepted; however, small, non-significant trends pointed to considerations for future design.

Overall, the results indicated that the 3D condition tended to support faster task completion and lower discomfort compared to the 2D condition. In addition, workload scores were slightly lower in the 3D visualization, whereas the 2D visualization showed a slight advantage in usability. Memory recall results suggested similar outcomes in both conditions, though missing responses limited firm conclusions. Along with the research gaps found during the scoping review, these findings contribute to a better understanding of how SV can support industrial training, and highlight the importance of choosing visualization methods that are clear, comfortable, and supportive of learning while balancing UX with task performance.

CRediT authorship contribution statement

Zeinab BagheriFard: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Miruna Maria Vasiliu:** Writing – review & editing, Validation, Investigation. **Emma Jane Pretty:** Writing – review & editing, Investigation. **Luis Quintero:** Writing – review & editing, Visualization, Supervision, Methodology. **Benjamin Edvinsson:** Writing – review & editing, Funding acquisition, Conceptualization. **Mario Romero:** Writing – review & editing, Funding acquisition. **Renan Guarese:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Benjamin Edvinsson reports a relationship with AstraZeneca AB that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.cag.2025.104500.

Data availability

The dataset used in this article is available at Mendeley Data, V1, doi: https://doi.org/10.17632/h5t39dkxf2.1.

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